IES · GenAI for Engineers

Lab 3.2
DMAIC Improvement Report

Goal	Turn six weeks of deviation data into a five-phase DMAIC summary with a Pareto chart.
You will produce	Phase rates with formulas shown, a Pareto chart, and a five-row DMAIC summary.
Time box	30 minutes
Outcome	LO1 explain DMAIC · LO2 automate the report with Copilot in Excel
Case	CW-401 chemical dosing skid

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.

Before you start

Open the Starter workbook for this lab. Do not open Worked-Reference.xlsx until you have finished your own attempt and checked it. Sign in to Microsoft 365 at https://m365.cloud.microsoft/ with your training account.

The concept in one minute

DMAIC is a roadmap that stops teams jumping to a fix. Define states the problem in measurable terms. Measure establishes a baseline you can trust. Analyse separates what the data shows from what someone believes. Improve tests a controlled change. Control keeps the gain. The discipline that matters most is in Analyse: a rate that fell after a change is an observation, not proof that the change caused it, especially over three weeks.

Step-by-step

Step 1 — Open the workbook and read the six weeks

Open **Starter.xlsx** and look at **Weekly_Data**.

Three weeks are Baseline and three are Pilot. Each week has Opportunities and Deviations. The Deviation rate column is empty and so is the phase summary below the table.

Step 2 — Understand the metric before you calculate it

Deviation rate = Deviations divided by Opportunities. It is a proportion, not a count.

This matters because the pilot weeks have roughly half the opportunities of the baseline weeks. Comparing raw counts would be wrong and would flatter the result.

Step 3 — Select WeeklyData and open Copilot

Click inside the table, confirm the name **WeeklyData**, then open Copilot in Excel.

Paste the prompt from **Lab-3.2-Prompt.pdf**. Note that it carries an E-MAP block: it fixes the units, states the assumption that the deviation definition never changed, and limits the scope to these six weeks only.

Step 4 — Verify both phase rates by hand

Do not accept the rates without checking. Baseline:

```
= 36 / 400 = 0.0900 = 9.00%
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Pilot:

```
= 7 / 200 = 0.0350 = 3.50%
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The absolute change is 5.50 percentage points. If Copilot reports "a 61% reduction", that is a relative change and it is misleading in a report. Ask for percentage points.

Step 5 — Build the Pareto

Go to the **Pareto** sheet. The five categories sum to 36, matching the baseline deviations.

Ask Copilot to add a cumulative percentage column and insert the Pareto chart.

Check that the top two categories account for most of the total. That is what tells the team where to look first.

Step 6 — Draft the five DMAIC rows

Ask for the five-row summary: Define, Measure, Analyse, Improve, Control.

Read the Analyse row hardest. Every causal statement must be labelled as a hypothesis unless the data in front of you supports it.

Step 7 — Force the limitation into the output

If Copilot did not say it, ask directly: what does this data NOT prove?

The answer must include that a three-week pilot cannot demonstrate a sustained improvement, and that if two changes were made at once they cannot be separately attributed.

Paste that limitation into the Control row. A report without it is not review-ready.

Step 8 — Compare with the worked reference

Open **Worked-Reference.xlsx** and check your rates and cumulative percentages match.

Check your own work

Do these by hand before you accept the Copilot output. If a number disagrees, the Copilot answer is wrong until proven otherwise.

- Baseline rate by hand: 36 / 400 = 9.00%.
- Pilot rate by hand: 7 / 200 = 3.50%.
- Absolute change is 5.50 percentage points, stated as percentage points not percent.
- Pareto counts sum to 36 and reconcile with the baseline deviations.
- The output states plainly that the short pilot does not prove cause.

Engineering boundary

Do not invent a root cause, a cost saving or a completion percentage. Do not present a relative improvement as though it were the absolute change. The final status is draft; a competent reviewer decides whether the pilot is adopted.

Evidence to retain

Save the final prompt, the workbook, one independent hand check, any correction you made and the name of the human reviewer.

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